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Chapter 26

GELFAND-P001199

9/IX–85 Gelfand seminar.

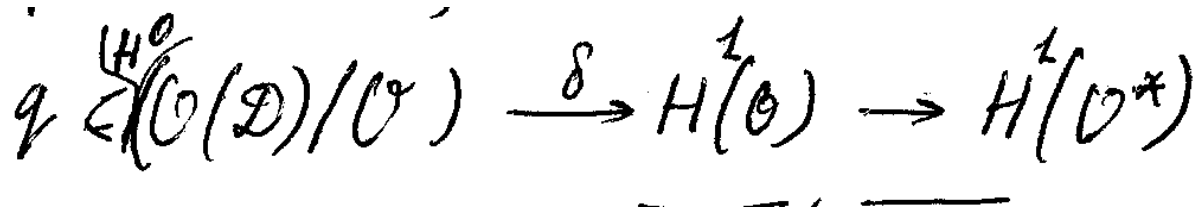
Kapranov, Misha.

Baker functions and their applications to the Yang–Mills equations.

Baker function of an arbitrary set — the simplest functions with essential singularities.

$$(1) \quad \varphi = e^{q(z)}\varphi_0, \qquad \varphi_0 \text{ — holomorphic}$$

$$(2) \quad q \in H^0(\mathcal{O}(D)/\mathcal{O}) \xrightarrow{\delta} H^1(\mathcal{O}) \longrightarrow H^1(\mathcal{O}^*)$$



Original source details: GELFAND-P001199

Uncertain reading. H^0 is inserted above the line, before the first parenthesis.

— · —

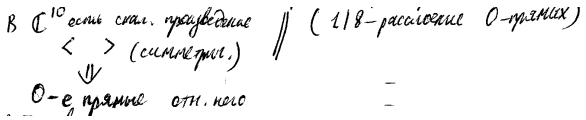
Now something inspired by Witten,

In $\mathbb{C}^{10|16}$ there is a family of lines of dimension $1/8$.



Original source details: GELFAND-P001199

Uncertain reading. The separator in the exponent $10 \mid 16$ is almost vertical. In the dimension $1/8$ it is slanted.



Original source details: GELFAND-P001199

Diagram and typed description.

In $\mathbb{C}^{10}$ there is an inner product $\langle \ \rangle$ (symmetric) $\Downarrow$ null lines with respect to it // (1|8-bundle of null lines)

[In the left margin:] A. Goncharov.

$SO(10)$ acts on $(\mathbb{C}^{10}, \langle\,,\,\rangle)$

$V_0 \oplus V_1$, V_1 — spinor representation of the group $SO(10)$.



Uncertain reading. The line about the spinor representation reads $SO(10)$.

Original source details: GELFAND-P001199

dim $V_1 = 16$

Subspaces of dim. $(1, 8)$: $L_0 + L_1$

L_0 — a null line in $\mathbb{C}^{10} \Longrightarrow$ subgroup $SO(8) \subset SO(10)$

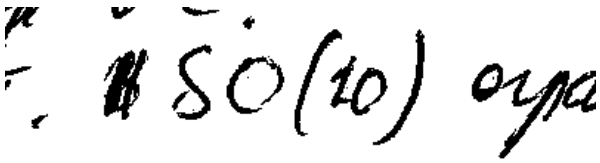
[Below the right-hand side:] (in the stabilizer of the line L_0)

(its semisimple part — $SO(8)$)

This is the group of rotations preserving L_0 and the orthogonal complement (which contains it)

$\Longrightarrow$ obtain rotations of the quotient space.

Now the spinor representation



Uncertain reading. The mark before $SO(10)$ is heavily overwritten. It may be the Russian preposition "в" (in).

Original source details: GELFAND-P001199

$SO(10)$ restrict to $SO(8)$; obtain 2 invariant subspaces (8-dimensional)

GELFAND-P001200

[source number: 59]

$\Longrightarrow$ 2 families of subspaces.

[Crossed out: illegible letters[?].]



Uncertain reading. illegible letters

Original source details: GELFAND-P001200

Consider the integrability condition for a connection along [inserted above: these] subspaces. This is a linear condition on the curvature, agreeing with the equations written earlier in a different context.

— · —

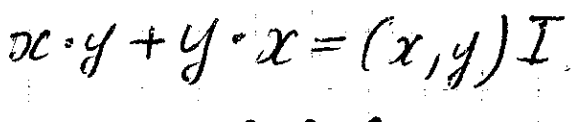
Spinor representation.

$$e_1, \ldots, e_{10}: \qquad e_i e_k + e_k e_i = 2\delta_{ik} I$$

and the ring they generate

[At right, beyond a double vertical stroke:]

$$x \cdot y + y \cdot x = (x, y) I$$



Uncertain reading. In the formula to the right of the stroke, $x \cdot y + y \cdot x = (x, y)I$ is written without a factor of 2.

Original source details: GELFAND-P001200

Dim. of the ring: $e_1, \dots, e_{10}$; $\quad e_i e_k, \quad e_i e_j e_k \qquad (e_i^2 = I)$

In all dim $= 2^{10}$; it is simple (— isomorphic to a matrix ring).

[Under "matrix":] $2^5 \times 2^5$

Obtain a representation of dimension 32.

$A \in SO(10)$ preserves the formulas $\Longrightarrow$ an automorphism of the matrix ring, and it is inner $\Longrightarrow$

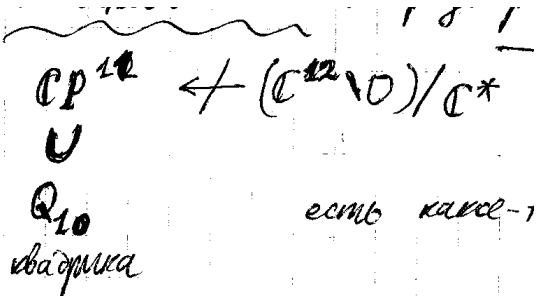
$\Longrightarrow$ a representation arises.

— · —

Curvature on the subspaces $= 0 \Longleftrightarrow$ generalized self-dual Yang–Mills equation.

Ordinary Y.-M.: dimensions 4 and 2.

— · —



Original source details: GELFAND-P001200

Diagram and typed description.

$\mathbb{CP}^{12} \longleftarrow (\mathbb{C}^{12} \setminus 0)/\mathbb{C}^*$

$\cup$

Q_{10}

quadric

The final exponent digit on $\mathbb{CP}$ is overwritten. The reading 12 is tentative, possibly 11. A diagonal stroke crosses the arrow.

[To the right of Q_{10}:] there is some covering superspace [source abbr.][?].

Uncertain reading. superspace [source abbr.]

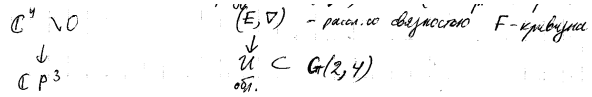


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— · —

Radon–Penrose transform.

$\mathbb{CP}^3$; lines in $\mathbb{CP}^3$ form $G(2,4)$ (2-dimensional subspaces of $\mathbb{C}^4$)



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Diagram and typed description.

$\mathbb{C}^4 \setminus 0$ $\qquad$ (E, ∇) — bundle with connection

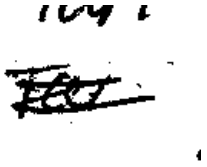
$\downarrow$ $\qquad$ $\downarrow$ F — curvature

$\mathbb{CP}^3$ $\qquad$ $U \subset G(2,4)$

region

α-planes in T_xU: there is a quadratic cone defined by the inner product.

In it [inserted above the line: 2] families of 2-dimensional [crossed out: illegible letters?]



Uncertain reading. illegible letters

Original source details: GELFAND-P001200

subspaces. Choose one of the families.

$F|_{\Pi_\alpha} = 0 \Longleftrightarrow$ the bundle is called self-dual.

$G(2,4)$ has rank 2 (first after spheres). To match a pair of points one needs to specify 2 numbers (one on a sphere).

Curvature $= 0 \Longrightarrow$ self-dual Y.-M. equation.

$\rightsquigarrow$ another bundle over $\mathbb{CP}^3$; α-surface — integral subset; the bundle is trivialized $\Longrightarrow$ it suffices to take the fibre over 1 point.

[At right, beyond a double vertical stroke:] Now a bundle without connection on another set.

GELFAND-P001201

V.I. Arnold. On [inserted above: (some)] special functions arising in bifurcation theory.

H — a polynomial

$H(x,y) = c$ — a family of algebraic curves (real)

$\omega = P\,dx + Q\,dy$ — a form (P, Q — polynomials)

$$\oint_{H=c} \omega = I(c)$$

How many zeros are there for the integral $I(c)$?

In one special case this is solved completely.

Theorem (German Petrov-Tan'kin?)



Uncertain reading. German Petrov-Tan'kin

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5th-year student, expelled? from the university



Uncertain reading. expelled

Original source details: GELFAND-P001201

$$H = \frac{y^2}{2} + \underset{\text{a polynomial}}{P_3(x)}$$

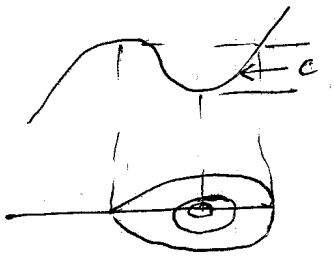
c varies in the interval with ovals?



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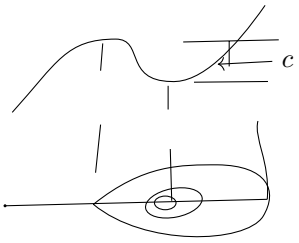
Uncertain reading. ovals

$\deg P \le n, \quad \deg Q \le n.$



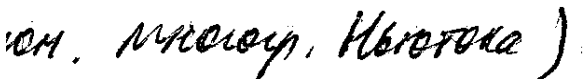
Original source details: GELFAND-P001201

Diagram and typed description.



The thin vertical strokes between the curves are broken in places.

$\dim\{I\} = n$ (count using the Newton polyh.?)



Original source details: GELFAND-P001201

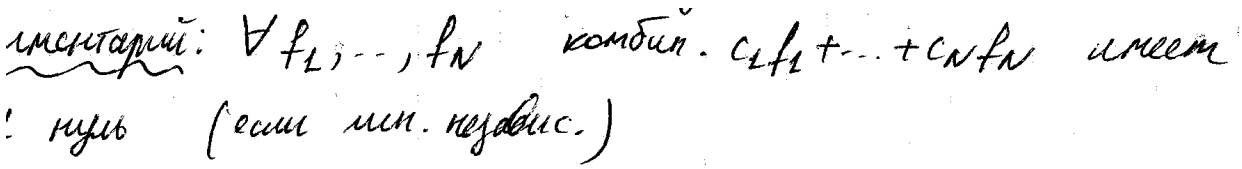
Uncertain reading. Newton polyh.

(the integrals are functions of c. They form a vector space since one can add the forms ω)

(!) Assertion: number of zeros $\overset{\text{of any function considered}}{<} n$ (if it is $\neq 0$)

the functions satisfy a Picard–Fuchs equation (Gauss–Manin connection)

Comment: $\forall f_1, \dots, f_N$ the combination $c_1 f_1 + \dots + c_N f_N$ has $N-1$ zeros (if linearly independent)

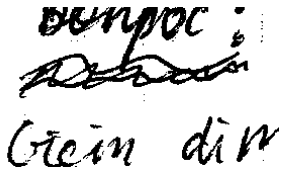


Original source details: GELFAND-P001201

Uncertain reading. The wording reads “has $N-1$ zeros”. No inequality sign is visible here.

If the number of zeros is $<$ dim, the system is called Chebyshev (nonoscillation)

Question: relation to oscillation theorems?



Original source details: GELFAND-P001201

Uncertain reading. A short line beneath “Question” is hatched over. No letters can be read with certainty.

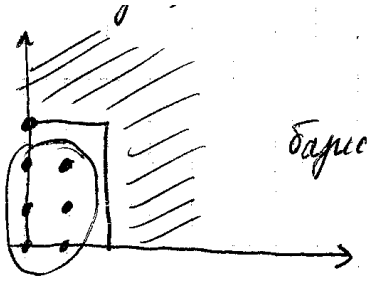
— · —

Counting dim: $\int = 0$ if a) an exact differential

b) multiply something by $H - c$.

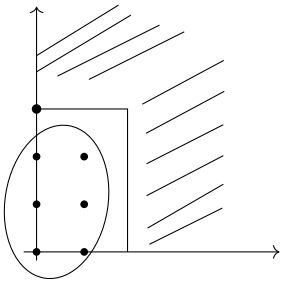
Using the Newton polyhedron to compute dimensions

$$f = x^3 + y^4; \qquad \mathbb{C}[[x,y]]/(\tfrac{\partial f}{\partial x}, \tfrac{\partial f}{\partial y})$$



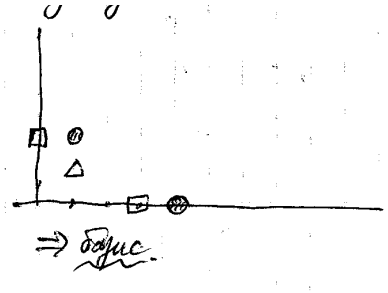
Original source details: GELFAND-P001201

Diagram and typed description.



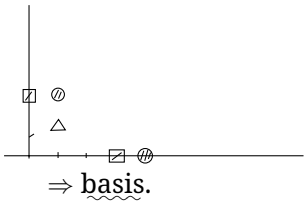
basis of the quotient space. dim $= 6$

$x^2y + y^4$



Original source details: GELFAND-P001201

Diagram and typed description.



The interior strokes in the squares and circles are schematic. Their reading remains uncertain.

GELFAND-P001202

[source number: 60]

A. Zelevinsky. On “the most beautiful result in pure math.”

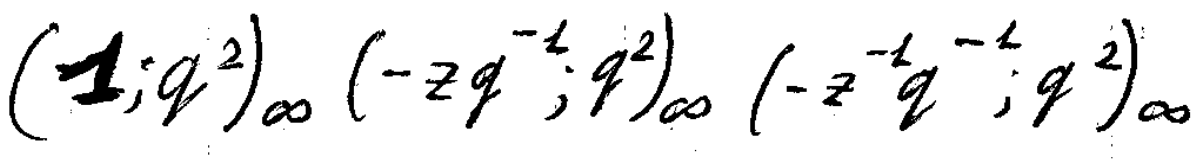
(Jacobi triple product)

$$(1) \qquad \sum_{r=-\infty}^{\infty} z^r q^{r^2} = \prod_{k=1}^{\infty} (1-q^{2k})(1+q^{2k-1}z)(1+q^{2k-1}z^{-1})$$

(Jacobi triple product)

$$\|$$

$$(1;q^2)_\infty (-zq^{-1};q^2)_\infty (-z^{-1}q^{-1};q^2)_\infty$$



Original source details: GELFAND-P001202

Uncertain reading. The first factor has 1 before the semicolon. The next two factors have q^{-1}.

Here

$$(a;q)_\infty := \prod_{n=0}^{\infty}(1-aq^n)$$

!! Follows from Ramanujan's identity.

Combinatorial proof

Move the first product to the left-hand side

$$(2)\qquad \left(\sum_{r=-\infty}^{\infty} z^r q^{r^2}\right)\prod_{k=1}^{\infty}(1-q^{2k})^{-1} = \prod_{k=1}^{\infty}(1+q^{2k-1}z)(1+q^{2k-1}z^{-1})$$



Uncertain reading. A mark above z^r on the left of (2) is heavily crossed out.

Original source details: GELFAND-P001202

Only the coefficient at the following type of monomial is nonzero $z^r q^{r^2+2n}$

(on the left and on the right)

$=$ number of partitions of n into summands $= p(n)$

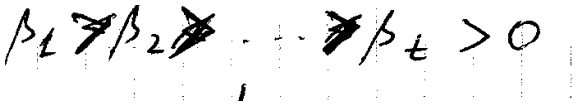
(on the left)

(3) On the right: exponents

$$\alpha_1 > \alpha_2 > \cdots > \alpha_s > 0$$

odd

$$\beta_1 > \beta_2 > \cdots > \beta_t > 0$$

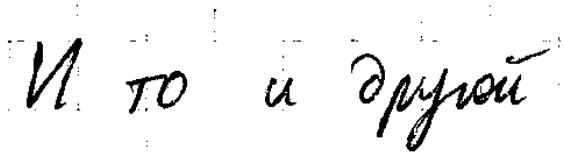


Uncertain reading. The three comparisons between the β terms are heavily retraced and read as $>$.

Original source details: GELFAND-P001202

$$\sum \alpha_j + \sum \beta_i = r^2 + 2n, \qquad s - t = r$$

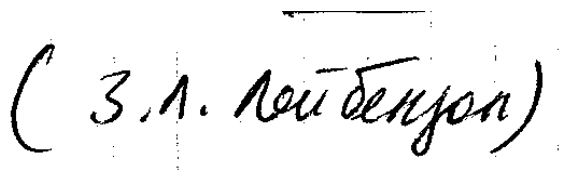
[Two diverging strokes from "odd" point toward the α and β lines. Vertical strokes flank the lines.]



Uncertain reading. Both the one and the other [irregular phrasing in the manuscript]

Original source details: GELFAND-P001202

(Z.L. Leibenzon[?]) — combinatorial description of the set:



Uncertain reading. Z.L. Leibenzon

Original source details: GELFAND-P001202

$00000\,1\,0000\,1\,00\,111111$

infinite words (linkages)

[Crossed out: on the left[?]]



Uncertain reading. on the left

Original source details: GELFAND-P001202

first infinitely many 0; then a jumble; then strokes

(infinitely many)

Height: $h(A)$ — number of inversions $\underbrace{1\,0}$ — arbitrarily far apart

$h(A) = n \quad \Rightarrow$ number of elements $p(n)$

— parametrized by tuples of the form (3).

1st method

$0000\,1\,0\,111$ — height 1

$p(n)$ — take the leftmost stroke and count zeros after it, and so on.

2nd method

The gaps have a [above "number": (natural)] number: (number of 1 on the left) $-$ (number of 0 on the right)

The number varies from $-\infty$ to $+\infty$!! (throughout!)

GELFAND-P001203

Arnold's remark:

Let us draw:

0 — a step to the right

1 — a step up

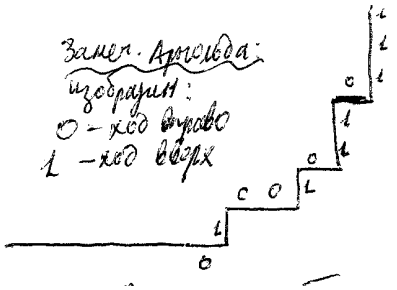
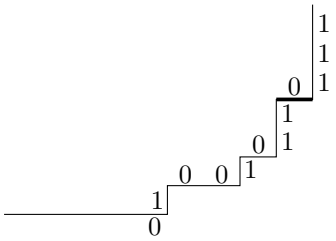


Diagram and typed description.



Original source details: GELFAND-P001203

Young diagram

the function $x - y$ numbers all

Consider the gap numbered r.

We characterize the linkage as follows:

the distance from the gap to the sticks, *[beneath the word, with a brace: lying to the left]* || odd numbers.

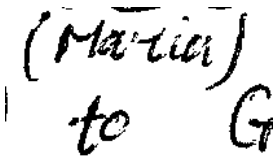
— " — the circles. *[beneath the word, with a brace: lying to the right]*

$$\sum \alpha_j + \sum \beta_j = r^2 + 2n$$

from the area under the curve.

16/IX–85 Gelfand Seminar.

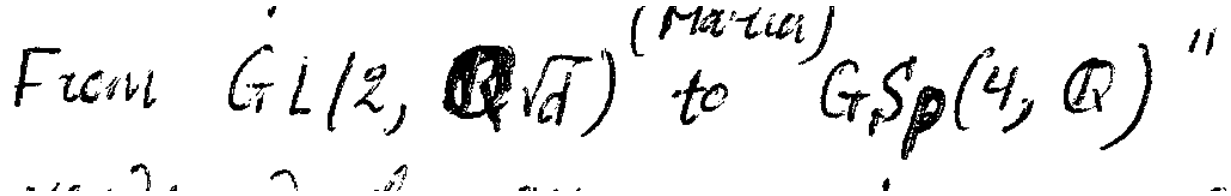
Zelevinsky (and Beilinson) as reported by M.-F. Vigneras (Marie-France Vigneras)



Uncertain reading. Parenthetical note below the start of the name, provisionally read as "Ma-rie."

Original source details: GELFAND-P001203

1) on the work "From $\mathrm{GL}(2, \mathbb{Q}\sqrt{d})$ to $\mathrm{GSp}(4, \mathbb{Q})$" (M.-F. Vigneras)



Original source details: GELFAND-P001203

Uncertain reading. The first letter in the field notation is overwritten and is read as $\mathbb{Q}\sqrt{d}$.

Construction of a correspondence between representations.

(lifting)

On passing to a p-adic field

this makes it possible to verify for $\mathrm{GSp}(4, F)$ some Langlands conjecture on functoriality.

[At right, with a pointer to GSp*:]* symplectic similitudes (preserve the symplectic form up to a scalar)

[Below F:] p-adic

2) Rallis, Piatetski-Shapiro and others

$$L(s, \pi_1 \times \pi_2 \times \pi_3), \quad \text{where } \pi_j \text{ — representations of } \mathrm{GL}_2(F)$$

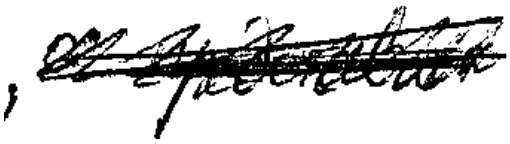
3) According to Jacket "How to use the trace formula?"



Uncertain reading. The surname in the manuscript reads "Jacket."

Original source details: GELFAND-P001203

The latest important result is the Paley–Wiener theorem,



Original source details: GELFAND-P001203

Uncertain reading. Crossed out after the theorem name: “on ....”

proved by Kazhdan and Bernstein.



Original source details: GELFAND-P001203

Uncertain reading. The abbreviation at the start of the line is read as “latest.”

[In an oval linked to the name of the result:] on representations over F

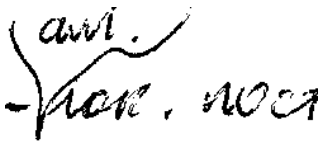
G — a reductive p-adic group (e.g. GL_n)

$H^\bullet(G)$ — locally constant, compactly supported functions on G.



Original source details: GELFAND-P001203

Uncertain reading. The raised mark after H is read as a dot.



Original source details: GELFAND-P001203

Uncertain reading. “Alg.” is inserted above the dash, with a curved mark linking the insertion to the line.

[To the right of the vertical stroke:] locally constant functions take the place of C^∞ functions

From f we construct:

$$\pi \longrightarrow \operatorname{tr} \pi(f)$$

— a function on $\operatorname{Irr} G$

Here

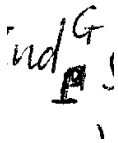
$$\pi(f) = \int_G f(g)\pi(g)\,dg$$

The theorem (Paley–Wiener, according to K.-B.) describes all such functions

Irreducible representations divide into sheets and families: from $\rho \in \operatorname{Irr} M$ one constructs $M \subset G$

$$\pi \in \mathrm{JH}\ \operatorname{Ind}_{\underline{P}}^{G} \rho$$

(Jordan–Hölder series)



Original source details: GELFAND-P001203

Uncertain reading. The overwritten lower index on induction is read as an underlined P.

‖ each family acquires a structure

M — the Levi subgroup of a parabolic subgroup of G

(semisimple part)

character of an irreducible representation $\Leftrightarrow$ algebraicity of a certain function

Idea in the real *[below "real," circled: and complex?]* case: decay on $\forall$ Cartan subalgebra



Original source details: GELFAND-P001203

Uncertain reading. The abbreviation is read as "decay."

GELFAND-P001204

[source number: 61]

M.-F. Vigneras. (on the same topic, but more precisely)



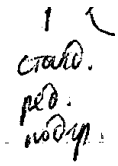
Original source details: GELFAND-P001204

Uncertain reading. The last word of the note is read as the abbreviation for "precisely."

G reductive p-adic group (e.g., GL_n)

$$\coprod_{(M,\rho)} \{\rho\chi\}^W = \operatorname{Irr} G \tag{1}$$

[Labels with pointers:] M — standard reductive subgroup, ρ — irreducible cuspidal representation, χ — unramified character, W — up to the action of the Weyl group.



Original source details: GELFAND-P001204

Uncertain reading. The first line of the label for M is read as the abbreviation for "standard."

$\{\rho\chi\}^W$ — finite-dimensional algebraic set (over $\mathbb{C}$).

from $f \in H(G)$ one constructs a function on [inserted: the set (1)]

$$\pi \longrightarrow \operatorname{trace} \operatorname{ind}_M^G \rho\chi(f)$$

Conditions

a) algebraic on $\forall$ such set

b) supported on finitely many of them

Beilinson.



Original source details: GELFAND-P001204

Uncertain reading. The words below the heading are crossed out and cannot be read with certainty.

Representations of $\mathrm{GSp}(4,\mathbb{Q})$ — ?

$$W_{\mathbb{Q}_p} \subset \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$$

$$\mathrm{Irr}\,\mathrm{GSp}(4,\mathbb{Q}) \longrightarrow \text{representations of the Weil–Deligne group}$$



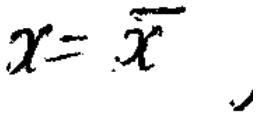
Uncertain reading. Unresolved crossed-out mark after the arrow, before "representations."

Original source details: GELFAND-P001204

[With a pointer to the right-hand side of the arrow:] the inverse image of $\forall$ point is finite (L packet)

$$\mathrm{GL}(2,\mathbb{Q}(\sqrt{d})) \longrightarrow \mathrm{GO}(4)$$

$$\left\{x \in M(2,\mathbb{Q}(\sqrt{d})) : x = \bar{x}\right\} \qquad x = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$$



Uncertain reading. The condition reads $x = \bar{x}$, with no visible transpose sign.

Original source details: GELFAND-P001204

det — canonical quadratic form (it is rational)

$$x \longmapsto gx\bar{g}^t \qquad \text{preserves this condition}$$

(analogue of the spinor representation)

Weil representation: representation of $\mathrm{GO}(4) \times \mathrm{GSp}(4)$

it is realized in θ functions

B. Feigin. On strings



Uncertain reading. The crossed-out word before "String theory" remains unresolved.

Original source details: GELFAND-P001204

String theory is the statistics of surfaces.

Quantum mechanics studies a space with a Lagrangian. The action depends on the trajectory.

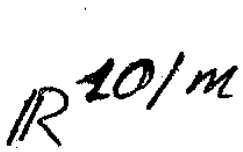
Strings: a surface takes the place of a trajectory.

$\mathcal{E}$ — a Riemann surface; consider the space $\mathbb{R}^d$

A trajectory is a map

$$\{\mathcal{E} \underset{x_1,\ldots,x_d}{\longrightarrow} \mathbb{R}^d\}.$$

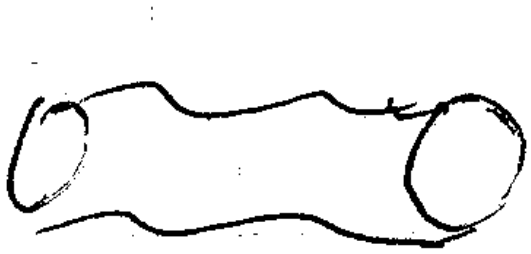
Idea: all elementary particles are described by maps from $\mathcal{E}$ to the superspace $\mathbb{R}^{10/m}$



Original source details: GELFAND-P001204

Uncertain reading. The exponent shows a slash and m: $10/m$.

The motion of a circle gives a surface



Original source details: GELFAND-P001204

Diagram and typed description.



The classical analogue is crystal growth.

GELFAND-P001205

Lagrangian:

$$\mathcal{L} = \int [\text{crossed out}]\,(\det g_{ij})^{1/2}\,d\sigma\,d\tau \qquad \text{(Nambu Lagrangian)} \tag{1}$$



Original source details: GELFAND-P001205

Uncertain reading. The passage crossed out immediately after the integral sign remains unresolved.

(area in the induced metric)

We need to consider the Feynman integral.

$$X : \mathcal{E} \hookrightarrow \mathbb{R}^d$$

Another way to write (1):

$$\mathcal{L}(X) = \int d\sigma\, d\tau \left[\left(\frac{\partial X}{\partial \sigma}\right)^2 \left(\frac{\partial X}{\partial \tau}\right)^2 - \left(\frac{\partial X}{\partial \sigma} \cdot \frac{\partial X}{\partial \tau}\right)^2 \right]^{1/2} \tag{1'}$$

Feynman integral

$$\int e^{\beta \mathcal{L}(X)}\, \mathcal{D}X \tag{2}$$

There are two approaches: via integrals and via ordinary operators (!). A correspondence has now been established.

Polyakov: another continuum integral

$\widetilde{X}$ — [inserted: a space] whose elements are pairs

1) $X : \mathcal{E} \longrightarrow \mathbb{R}^d$

2) g_{ij} on $\mathcal{E}$ (a Riemannian metric on $\mathcal{E}$)

Polyakov Lagrangian:

$$\text{[crossed out]}\ \mathcal{L} = \text{æ} \int (\det g_{ij})^{1/2} + \int \frac{\partial X_i}{\partial \xi_\alpha} \frac{\partial X_i}{\partial \xi_\beta} g^{\alpha\beta}\, \overset{\text{[inserted]}}{\sqrt{g}}\, d\xi_1\, d\xi_2 \tag{3}$$



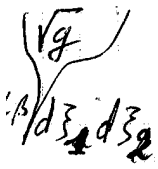
Original source details: GELFAND-P001205

Uncertain reading. The crossed-out mark before $\mathcal{L}$ remains unresolved.



Original source details: GELFAND-P001205

Uncertain reading. The coefficient before the first integral resembles “æ.”



Original source details: GELFAND-P001205

Uncertain reading. $\sqrt{g}$ is inserted above the line. The overwritten subscripts in $d\xi_1\, d\xi_2$ are read as $1, 2$.

[Inserted: To it] the calculation technique applies

Procedure: fix g_{ij} and evaluate the integral over X — it is quadratic.

— · —

GELFAND-P001206

[source number: 62]

23.9.85 Gelfand Seminar.

I.M.G. Problem (for grade 9) What polyhedron results from cutting a 4-dimensional cube by a 3-dimensional hyperplane passing through the center and perpendicular to the main diagonal?

A. Beilinson. On a new series of papers by Piatetski-Shapiro and Rallis.

functional equations of L-functions associated with automorphic forms

G — a semisimple group over $\mathbb{Q}$ (e.g., the group of matrices $n \times n$)

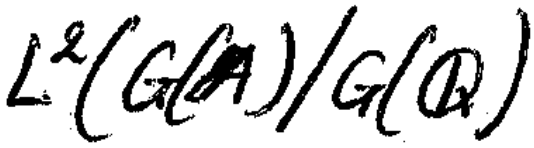
$G/\mathbb{Q}$.



Uncertain reading. The signs to the right of $G/\mathbb{Q}$ are heavily crossed out.

Original source details: GELFAND-P001206

Problem: describe the representations of the adelic group $G(A)$ in $L^2(G(A)/G(\mathbb{Q})$



Uncertain reading. The notation for the L^2 space has one closing parenthesis after $\mathbb{Q}$.

Original source details: GELFAND-P001206

The corresponding representations are called automorphic.

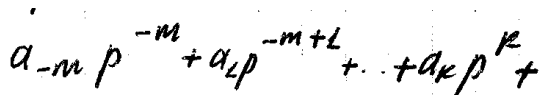
— · —

I.M.G. $\forall$ natural number can be written in the form

$$a_0 + a_1 p + a_2 p^2 + \ldots + a_n p^n$$

Any rational number

$$a_{-m} p^{-m} + a_1 p^{-m+1} + \ldots + a_k p^k +$$



Uncertain reading. The second term reads $a_1 p^{-m+1}$. A $+$ follows $a_k p^k$.

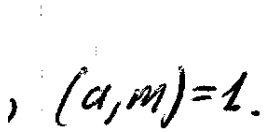
Original source details: GELFAND-P001206

For example,

$$\frac{1}{3} = 1 + 1 \cdot 2 + 1 \cdot 2^3 + 1 \cdot 2^5 + \dots$$

[At right:]

$$\|a\| = p^{-n}, \quad \text{if } a = p^n m, \quad (a, m) = 1.$$

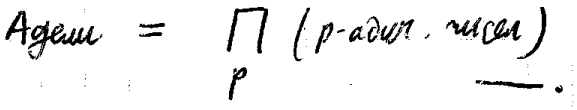


Uncertain reading. The coprimality condition reads $(a, m) = 1$.

Original source details: GELFAND-P001206

$$\underline{p = 2} \quad \Rightarrow \quad \|2\| = \frac{1}{2}; \quad \|3\| = 1, \quad \|4\| = \frac{1}{4}, \quad \left\|\frac{1}{2}\right\| = 2.$$

$$\text{Adeles} = \prod_p (p\text{-adic numbers})$$



Uncertain reading. The p appears below the product sign.

Original source details: GELFAND-P001206

— · —

$$\mathrm{SL}_2(\mathbb{R}) \supset \mathrm{SL}_2(\mathbb{Z}) \supset \Gamma_n$$

[Label pointing to Γ_n:] matrices of order 2, [crossed out: elem.?] that are congruent to 1 (matrix) mod$n$.



Uncertain reading. The abbreviation crossed out after "order" may read "elem."

Original source details: GELFAND-P001206



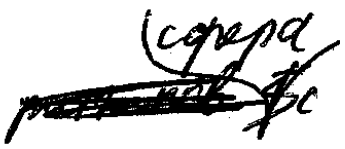
Uncertain reading. The ending of the word "congruent" is overwritten.

Original source details: GELFAND-P001206

$$\Gamma_{n_1} \supset \Gamma_{n_2} \quad \text{when } n_1 \mid n_2$$

$$G(A)/G(\mathbb{Q}) = \varprojlim_n G(\mathbb{R})/\Gamma_n.$$

Γ_n acts on the upper half-plane H; H/Γ_n — [crossed out: Riemann?] [inserted: a sphere] (with punctures)



Uncertain reading. The first crossed-out word may be "Riemann." "A sphere" is inserted above the line.

Original source details: GELFAND-P001206

$H/\mathrm{SL}_2(\mathbb{Z})$ — a sphere with two punctures

H/Γ_2 — — ″ — three punctures



Original source details: GELFAND-P001206

Uncertain reading. The first line says “a sphere with two punctures.”

$$\pi_{G(A)} = \pi_{G(\mathbb{R})} \otimes \pi_{G(\mathbb{Q}_2)} \otimes \dots$$

— · —

I.M.G. At Sem. Bourbaki there is a paper by A. Connes on the relation between the topology of knots and invariants of factors!!



Uncertain reading. The abbreviation before “knots” reads “top.”

Original source details: GELFAND-P001206

— · —

It is enough to know the representations of the p-adic group.



Uncertain reading. The crossed-out mark at the end of the line remains unresolved.

Original source details: GELFAND-P001206

Which occur in $L^2(G(A)/G(\mathbb{Q}))$? Far from all of them.

Representations of $G(\mathbb{Q}_p)$.

The simplest are spherical, when there is a vector invariant under a [inserted: max] compact subgroup.

$$G(\mathbb{Q}_p)/B(\mathbb{Q}_p)$$

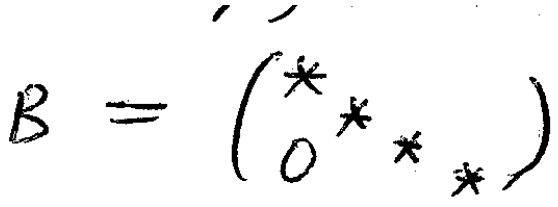
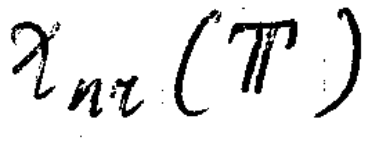


Diagram and typed description.

$$B = \begin{pmatrix} * & * & * \\ 0 & * & \\ & & * \end{pmatrix}$$

Original source details: GELFAND-P001206

are indexed by unramified characters of the torus $\chi_{nr}(T)$ = points of the dual torus $= T^0(\mathbb{C})$ (complex points)



Original source details: GELFAND-P001207

Uncertain reading. The subscript on χ is read as nr.

(trivial on the intersection of the torus with the maximal compact subgroup)

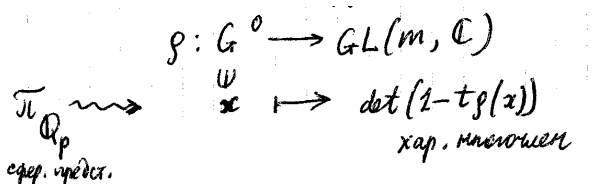
Spherical representations of $G(\mathbb{Q}_p) = T^0(\mathbb{C})/W$

unramified character $\longrightarrow$ conjugacy class in the matrix group

— · —

Let G — a split group (the maximal torus splits)

G^0 — dual group $\quad (G = \mathrm{GL} \Rightarrow G^0 = \mathrm{GL})$



Original source details: GELFAND-P001207

Diagram and typed description.

$$\begin{array}{ccccc} & & \rho: G^0 & \longrightarrow & \mathrm{GL}(m,\mathbb{C}) \\ & & \cup & & \\ \pi_{\mathbb{Q}_p} & \rightsquigarrow & x & \longmapsto & \det(1-t\rho(x)) \\ \text{spherical rep.} & & & & \text{characteristic polynomial} \end{array}$$

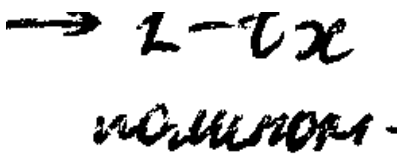
Suppose that for $\pi_{G(A)}$ almost all representations [Inserted above the line: (of the local groups $G(\mathbb{Q}_p)$)] are spherical (all but finitely many)

$$L(\pi,\rho,s) = \prod_p \frac{1}{\det\left(1 - p^{-s}\rho\left(\begin{array}{c} x_{\pi_{\mathbb{Q}_p}} \\ | \\ \text{semisimple element} \end{array}\right)\right)}$$

— · —

Example.

$$\rho: \mathrm{GL}_1 \underset{\text{identity}}{\longrightarrow} \mathrm{GL}_1 \quad \Rightarrow \quad x \longmapsto \underset{\text{polynomial}}{1-tx}$$



Original source details: GELFAND-P001207

Uncertain reading. The abbreviation below $1-tx$ is read as “polynomial.”

Uncertain reading. The crossed-out mark before $x = 1$ remains unresolved.



Original source details: GELFAND-P001207

$x = 1 \Rightarrow L(\pi, \rho, s) = \prod_p \frac{1}{1-p^{-s}}$ — the Riemann ζ-function.

Feigin B. On strings (continued).

$\mathcal{E}$ — a Riemann surface (a 2-dimensional set); $\mathbb{R}^d$ — a space with the Minkowski metric $\langle\ ,\ \rangle$ (1 negative square) or the Euclidean metric.

Consider the space [Inserted above the line: X] of smooth maps $\mathcal{E} \xrightarrow{x} \mathbb{R}^d$

[Inserted above the line: The map x induces] on $\mathcal{E}$ [crossed out] a Riemannian metric.

Uncertain reading. The crossed-out word after $\mathcal{E}$ may start with the Russian letters “зад...”.



Original source details: GELFAND-P001207

Let ξ_1, ξ_2 — coordinates [crossed out] on $\mathcal{E}$.

Uncertain reading. The crossed-out writing between “coordinates” and “on” remains unresolved.



Original source details: GELFAND-P001207

$$g_{ij} = \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j}\right) \qquad \{g^{ij}\} = \{g_{ij}\}^{-1} \tag{1}$$

Nambu Lagrangian:

$$\mathcal{L}(x) = \int_{\mathcal{E}} \omega = \int_{\mathcal{E}} \sqrt{g}\, d\xi_1\, d\xi_2, \qquad g = \det\{g_{ij}\} \tag{2}$$

(area of the surface in the Riemannian metric)

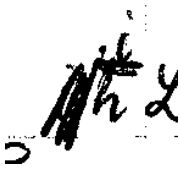
GELFAND-P001208

[source number: 63]

Path integral

$$\int e^{\frac{i}{h}\mathcal{L}(x)} \mathcal{D}x \tag{3}$$

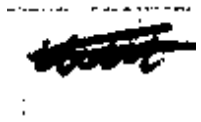
Uncertain reading. The overwritten coefficient in the exponent is read as i/h. The crossed-out mark on its left remains unresolved.



Original source details: GELFAND-P001208

Over all surfaces passing through given points [crossed out] etc. (to be specified below).

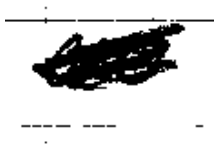
Uncertain reading. The word crossed out after "points" remains unresolved.



Original source details: GELFAND-P001208

We want to find the [crossed out] min of the Lagrangian $\mathcal{L}$.

Uncertain reading. The crossed-out mark before min remains unresolved.



Original source details: GELFAND-P001208

This is a minimal surface.

Properties of minimal surfaces.

a) $\mathcal{L}$ is independent of the choice of coordinates.

Variation of $\mathcal{L}$.

Let there be a 1-parameter family $x(t)$ of such surfaces.

$$\left.\frac{\partial \mathcal{L}}{\partial t}\right|_{t=0} = -\int (\dot{x}_t, H)\sqrt{g}\, d\xi_1\, d\xi_2. \tag{4}$$

Here

$$H = \Delta x = \frac{1}{\sqrt{g}} \sum_i \frac{\partial}{\partial \xi_i} \left(\sqrt{g} \sum_j g^{ij} \frac{\partial x}{\partial \xi_j} \right)$$

(the Laplace operator of the metric)

(5) Corollary:

if $x : \mathcal{E} \longrightarrow \mathbb{R}^d$ is given by harmonic functions, the surface is minimal.

— · —

I.M.G. Eells studies minimal surfaces.

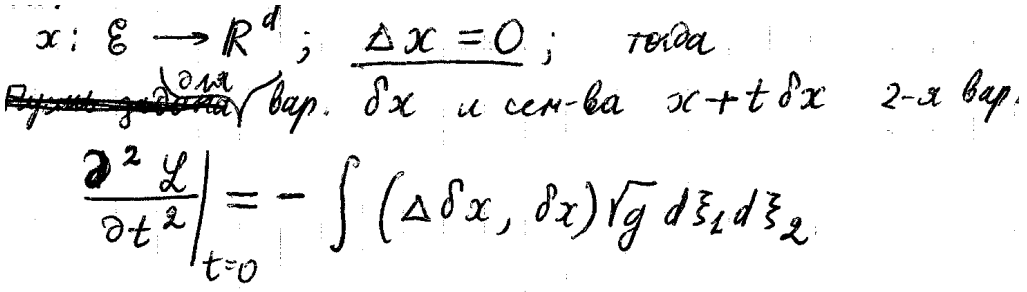
Lashof — " ——; which are constructed from groups.

Let;

$$x : \mathcal{E} \longrightarrow \mathbb{R}^d; \qquad \underline{\Delta x = 0;} \qquad \text{then} \tag{6}$$

[crossed out: Let there be] [Inserted above the line: For] a variation δx and the family $x + t\delta x$ the second variation takes the form

$$\left.\frac{\partial^2 \mathcal{L}}{\partial t^2}\right|_{t=0} = -\int (\Delta\delta x, \delta x)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{7}$$

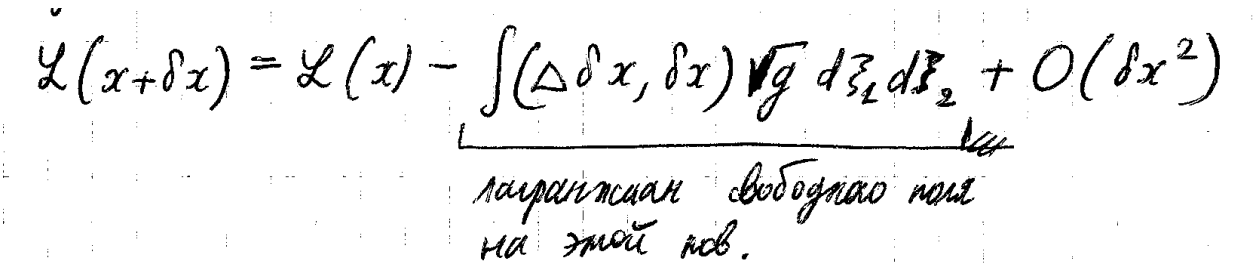


Original source details: GELFAND-P001208

Uncertain reading. The ∂ in the numerator is overwritten. The condition $\Delta x = 0$ appears above in (6).

We then obtain

$$\mathcal{L}(x + \delta x) = \mathcal{L}(x) - \underbrace{\int (\Delta\delta x, \delta x)\sqrt{g}\, d\xi_1\, d\xi_2}_{\text{Lagrangian of the free field on this surface}} + O(\delta x^2) \tag{8}$$



Original source details: GELFAND-P001208

Uncertain reading. The manuscript shows no factor $1/2$ before the integral and writes the remainder $O(\delta x^2)$.

Now the metric g_{ij} is fixed on $\mathcal{E}$. Replacing δx with x, we write the free-field Lagrangian in d form.



Uncertain reading. The mark before the Russian word for “form” resembles d.

Original source details: GELFAND-P001208

$$\mathcal{L} = \int \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j} \right) g^{ij} \sqrt{g}\, d\xi_1\, d\xi_2 \tag{9}$$

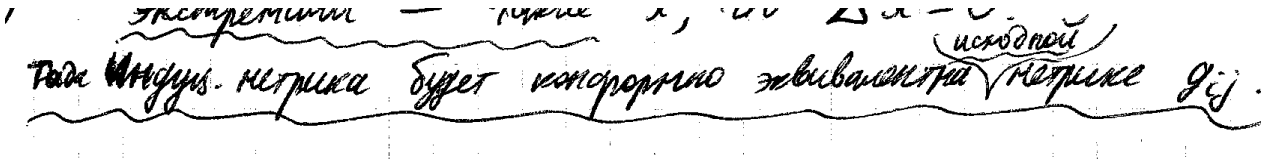
(10) The extremals are those x, for which $\Delta x = 0$.

(11) then the induced metric will be conformally equivalent to the [Inserted above the line: original] metric g_{ij}.



Uncertain reading. The underlined line begins with “Then.”

Original source details: GELFAND-P001208



Original source details: GELFAND-P001208

Uncertain reading. “Original” is inserted above the line, before “metric.”

GELFAND-P001209

Let us rewrite this:

$$\mathcal{L}(x, g_{ij}) = \int \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j} \right) g^{ij} \sqrt{g}\, d\xi_1\, d\xi_2 \tag{9'}$$

Consider the path integral

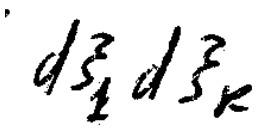
$$\int e^{ih^{-1}\mathcal{L}(x, g_{ij})} \mathcal{D}x\, \mathcal{D}g_{ij}. \tag{12}$$

Now x and g_{ij} are varied independently.

Let $\mathcal{D}$ — be the space of metrics. We give it a Riemannian metric

$$g_{ij} \longmapsto g_{ij} + \theta_{ij}$$

$$|\theta_{ij}|^2 = \int \theta_{ij} \theta_{k\ell} g^{ik} g^{j\ell} \sqrt{g}\, d\xi_1\, d\xi_k \tag{13}$$



Uncertain reading. The last differential in (13) reads $d\xi_k$.

Original source details: GELFAND-P001209

In (12) $\mathcal{D}g_{ij}$ — the (formal) volume form constructed from the metric (13).

The integral (12) [Inserted above the line: (undefined, since the integrand)] is invariant under the diffeomorphism group of the surface itself $\mathcal{E}$



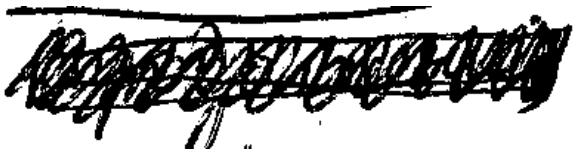
Uncertain reading. The surface symbol is read as $\mathcal{E}$, as on the preceding pages.

Original source details: GELFAND-P001209

$\Rightarrow$ constant on the orbits and undefined. One needs to use the Faddeev–Popov trick.

GELFAND-P001210

30/IX–85. B. Feigin (Gelfand Seminar)



Original source details: GELFAND-P001210

Undeciphered passage. The crossed-out line below the speaker's name remains undeciphered.

On Polyakov's work.

[I arrived $\frac{1}{2}$ an hour late]

Tools for calculating det.

$\mathcal{E}$ — a surface with a complex structure and a metric g_{ij}, compatible with the complex structure *[below the word "surface": (2-dimensional)]*.

T — the tangent bundle on $\mathcal{E}$, T^j — its jth tensor power

$$T^j \xrightarrow{\bar\partial} T^j \otimes \Omega^{0,1}, \qquad \Delta = \bar\partial^* \bar\partial$$

$$\zeta(s) = \sum \lambda_i^{-s}, \qquad \ln \operatorname{Det} \Delta = -\zeta'(0).$$

Det depends on g_{ij}

$$g_{ij} \rightsquigarrow g_{ij}(1+\varepsilon\rho) \qquad \text{(conformal change)}$$

$$\operatorname{Det}(\Delta + \varepsilon\delta\Delta) = \operatorname{Det}\Delta \cdot \operatorname{Det}(1 + \varepsilon\delta\Delta/\Delta)$$

$$\delta(\ln \operatorname{Det}\Delta) = \operatorname{Tr}(\delta\Delta/\Delta)$$

$\bar\partial$ does not depend on ρ; $\bar\partial^*$ depends on ρ

$$\begin{aligned}\Delta + \varepsilon\delta\Delta &= (1+\varepsilon\rho)^j \bar\partial^* (1+\varepsilon\rho)^{-j-1} \bar\partial \\ &= \bar\partial^*\bar\partial + \varepsilon \underbrace{\left(j\rho\bar\partial^*\bar\partial - (j+1)\bar\partial^*\rho\bar\partial\right)}_{\delta\Delta}\end{aligned}$$

Divide the second term by Δ.

$$\bar\partial^*\rho\bar\partial\bar\partial^{-1}(\bar\partial^*)^{-1} = \bar\partial^*\rho(\bar\partial^*)^{-1}$$

ρ is replaced by $\rho e^{-t\Delta}$ and we write $\operatorname{Tr}(\rho e^{-t\Delta})$ and then $t \to +0$; we have an expansion in powers of t

$$\operatorname{Tr}(\rho e^{-t\Delta}) = a_{-1}t^{-1} + a_0 + \dots \qquad \text{(2-dimensional surface)}$$

(see the standard textbooks: McKean and Singer)



Original source details: GELFAND-P001210

Uncertain reading. An undeciphered mark after the closing parenthesis is crossed out.



Original source details: GELFAND-P001210

Uncertain reading. Note below the names: "article in 'Matem.'"

a_{-1} — a linear functional of ρ; one can take the ρ, for which $a_{-1}(\rho) = 0 \Rightarrow$ for a subspace of codimension 1 it is well defined.

GELFAND-P001211

[source number: 66]

Otherwise

$$\operatorname{tr} \rho = æ(a_{-1}(\rho)) + a_0(\rho)$$

Uncertain reading. Here and below, the looped symbol most closely resembles “æ.”



Original source details: GELFAND-P001211

$a_{-1}t^{-1}$ — the infinite term.

All reasonable methods lead to this

(this follows from linearity and agreement where $a_{-1}(\rho) = 0$)

æ may depend on the metric.

The definition of Det is related to the definition of the trace.

— · —

Let Δ act on functions. Then

$$\operatorname{Tr}(\rho e^{-t\Delta}) = \frac{1}{4\pi t}\left(\int \rho\sqrt{g}\, d\xi_1\, d\xi_2 + \frac{t}{3}\int \rho K\sqrt{g}\, d\xi_1\, d\xi_2 + \dots\right)$$

[Below K:] (scalar curvature).

For $\Delta_j : T^j \to T^j$ we have:

$$\operatorname{Tr}(\rho e^{-t\Delta_j}) = \frac{1}{4\pi t}\left(2\int \rho\sqrt{g}\, d\xi_1\, d\xi_2 + \left(j + \frac{2}{3}\right) t \int \rho K\sqrt{g}\, d\xi_1\, d\xi_2\right)$$

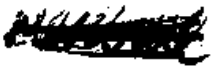
When $j = 0$ it is twice as [crossed out: small?] large, because we are in a complex bundle and take its realification.

Uncertain reading. The mark between j and 0 is overwritten and reads as equality.



Original source details: GELFAND-P001211

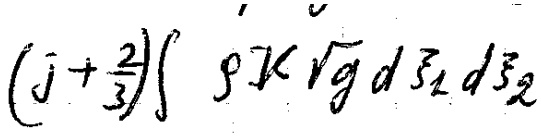
Uncertain reading. The crossed-out word before “large” may read “small.”



Original source details: GELFAND-P001211

Hence the trace of the operator of multiplication by ρ equals

$$\left(j+\frac{2}{3}\right)\int \rho K\sqrt{g}\,d\xi_1\,d\xi_2$$



Uncertain reading. The factor before the integral on this line is written as $(j+2/3)$.

Original source details: GELFAND-P001211

On $T^j\otimes\Omega^{0,1}$ we need to use $\Delta=\bar\partial\bar\partial^*$. This is the same as computing with

$$T^{-j-1}\longrightarrow T^{-j-1}\otimes\Omega^{0,1}$$

$$\operatorname{tr}(\rho\cdot e^{-t\bar\partial_j\bar\partial_j^*})=\operatorname{tr}(\rho e^{-t\Delta_{-j-1}})$$

$$\operatorname{Tr}\left(\frac{\delta\Delta}{\Delta}\right)=\frac{1}{4\pi}\frac{6j^2+6j+1}{3}\int\rho K\sqrt{g}\,d\xi_1\,d\xi_2+\text{æ}\int\rho\sqrt{g}\,d\xi_1\,d\xi_2$$

If the metric is $\theta(z,\bar z)\,dz\,d\bar z$, then $K=\partial\bar\partial\ln\theta$

Let $\theta(z,\bar z)=e^{\varphi}$

This gives a linear [Inserted above the line: of the variation] dependence on $\varphi\Rightarrow$ we integrate it and obtain



Uncertain reading. Above the word "dependence," "of the variation" is inserted.

Original source details: GELFAND-P001211

$$\ln\operatorname{Det}\Delta(\theta)=\mathcal{K}+\int\left(\partial\varphi\wedge\bar\partial\varphi+\text{æ}\,e^{\varphi}\,dz\,d\bar z\right)$$

[Below $\mathcal{K}$:] some number (it depends on everything!).

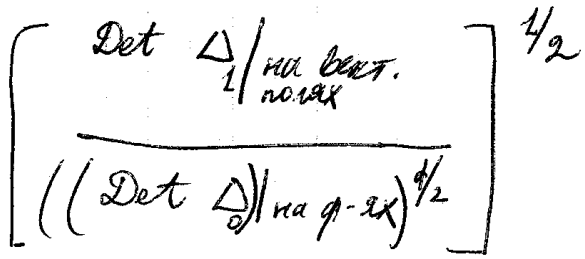
[At right:] (action of the Liouville equation).

We still need to be able to compute

$$\int e^{\mathcal{L}(\varphi)}\mathcal{D}\varphi,\qquad \mathcal{L}(\varphi)=\ln\operatorname{Det}\Delta$$

$\Leftrightarrow$ solve the Liouville equation.

$$\left[\frac{\operatorname{Det}\Delta_1|_{\text{on vector fields}}}{\left((\operatorname{Det}\Delta_0)|_{\text{on functions}}\right)^{d/2}}\right]^{1/2}$$



Uncertain reading. The denominator has exponent $d/2$, and the whole bracket has exponent $1/2$.

Original source details: GELFAND-P001212

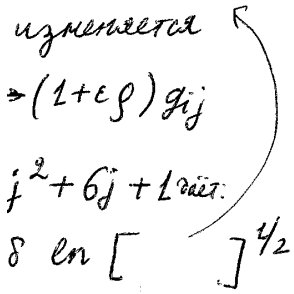
g_{ij} changes

$$g_{ij} \longrightarrow (1+\varepsilon\rho)g_{ij}$$

$6j^2+6j+1$ gives.

Hence [crossed out: ln?]

$$\delta \ln[\quad\quad]^{1/2} = \left(13-\frac{d}{2}\right)\int \rho K\sqrt{g}\,d\xi_1\,d\xi_2$$



Uncertain reading. The brackets are empty. A curved arrow leads from them to the word “changes.”

Original source details: GELFAND-P001212



Uncertain reading. The crossed-out writing before δ may read ln.

Original source details: GELFAND-P001212

This suggests $d = 26$

(these were maps of the surface into $\mathbb{R}^d$).

Then it is independent of the metric!

(if the metric is compatible with the complex structure!)

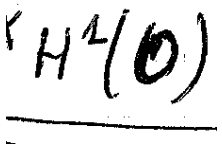
This is Polyakov's explanation of the role of 26 in string theory.

The constructions need to account for zero modes. We obtain

$$\mathcal{E} \rightsquigarrow \frac{\bigwedge^{\max} H^1(T)}{\bigwedge^{\max} H^1(O)}$$

[Below $\mathcal{E}$:] algebraic curve.

[Below the ratio, marked by a brace:] Here a canonical Hermitian metric arises.



Uncertain reading. The rounded sign in $H^1(O)$ resembles O or 0.

Original source details: GELFAND-P001212

GELFAND-P001213

14/X–85. Gelfand Seminar.

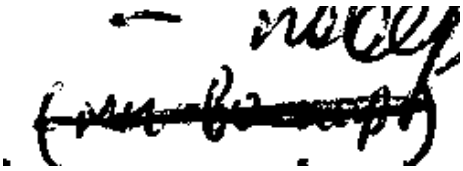
Fuchs — recapitulation [crossed out] [inserted above the line: previous] talk by Feigin



Uncertain reading. The crossed-out mark between "recapitulation" and "talk" may be I or 1.

Original source details: GELFAND-P001213

$\mathcal{E}$ — a surface (a 2-dimensional [inserted: smooth] compact [inserted: orientable] manifold without boundary)



Uncertain reading. The crossed-out parenthesis below the surface label may read "(set of pairs)."

Original source details: GELFAND-P001213

$$X = \{x : \mathcal{E} \to \mathbb{R}^d;\ g\}; \qquad g \text{ — a Riemannian metric on } \mathcal{E}$$

[Below X:] (X — the set of pairs).

$$\mathcal{L}(g, x) = \int_{\mathcal{E}} \left(\frac{\partial x}{\partial \xi_i}, \frac{\partial x}{\partial \xi_j} \right) g^{ij} \sqrt{g}\, d\xi_1\, d\xi_2 \tag{1}$$

I.M.G. Strings $\leftrightarrow$ 2 constants (not only h) in quantum mechanics,

Statistics of thin tubes instead of paths.

The second constant is the thickness of the tube.

GELFAND-P001214

[source number: 79]

F. Goal: find

$$\int_X e^{\frac{i}{h}\mathcal{L}(g,x)}\mathcal{D}g\,\mathcal{D}x \tag{2}$$



Uncertain reading. The overwritten label below the integral sign reads X.

Original source details: GELFAND-P001214

g — fixed

$$\int_{\{\mathcal{E}\to\mathbb{R}^1\}} e^{-\mathcal{L}(x,g)}\mathcal{D}x = (\mathrm{Det}\,\Delta)^{-1/2}, \tag{3}$$



Uncertain reading. The overwritten exponent on $\mathbb{R}$ in (3) reads 1.

Original source details: GELFAND-P001214

Δ — the Laplace–Beltrami operator of the metric g.

[Is there an analogue of Wiener measure for surfaces?]

$$\int_{\{\mathcal{E}\to\mathbb{R}^d\}} e^{-\mathcal{L}(x,g)}\mathcal{D}x = (\mathrm{Det}\,\Delta)^{-d/2} \tag{4}$$

We need

$$\int_{\{g\}} (\mathrm{Det}\,\Delta)^{-d/2}\mathcal{D}g \tag{5}$$

$\mathcal{M} = \{g\}$ — the space of metrics. Acting on it is Diff($\mathcal{E}$) [crossed out] and the function $(\mathrm{Det}\,\Delta)^{-d/2}$ is invariant under the action of this group.

Uncertain reading. The crossed-out mark before “and” may have been another “and.”



Original source details: GELFAND-P001214

$\mathcal{M}^c$ — the space of metrics conformally equivalent to the standard metric

(constant negative curvature; genus $g \geq 2$)

$$\mathcal{M}^c \times \mathrm{Diff}\,\mathcal{E} \xrightarrow{\text{onto}} \mathcal{M} \tag{6}$$

[At right:] (Theorem: $\forall$ metric can be made conformally equivalent to the standard one by a change of variables).

Faddeev–Popov trick

f — Diff-invariant function on $\mathcal{M} \Rightarrow$

$$\int_{\mathcal{M}} f\mathcal{D}g = \left(\int_{\mathcal{M}^c} f \cdot \nu\right) \cdot (\text{volume of the group Diff}) \tag{7}$$

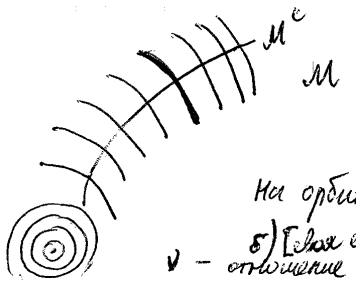
[In the right margin, beyond the double vertical line:] $\mathrm{vol}(\mathrm{Diff}) = \infty$, which we will later cross out.



Original source details: GELFAND-P001214

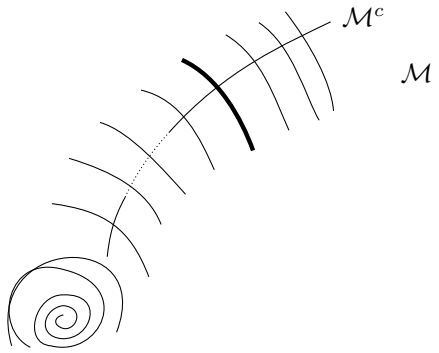
Uncertain reading. An undeciphered crossed-out mark is visible across the double line before “vol.”

Qualitative assertion: the main contribution comes from a small set (everything collapses to a point)



Original source details: GELFAND-P001214

Diagram and typed description.



The lower part of the sloping line is faint. The joins at the spiral are indistinct.

F.-P. trick

ν — the factor (at the integral over the orbits), which will be written out below.

There are two invariant metrics on the orbit; a) restriction of the metric of $\mathcal{M}$: dg

b) [its own natural metric (transferred from the group — the induced measure on G/H, H—the stabilizer subgroup)]

ν — the ratio of the two measures on each orbit.

GELFAND-P001215

δg — variation of the metric — tangent vector to the space of metrics.

$$|\delta g|^2 = \int_{\mathcal{E}} (\delta g, \delta g) \text{ vol. form} = \int_{\mathcal{E}} (\delta g, \delta g)\omega = \int_{\mathcal{E}} (\delta g, \delta g)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{8}$$

$\mathcal{D}g$ — the volume form [inserted: on $\mathcal{M}$], corresponding to $|\delta g|^2$ (9)

([overwritten mark] volume form constructed from the metric)



Uncertain reading. The overwritten mark before "volume form" may be an opening quotation mark or $=$.

Original source details: GELFAND-P001215

L — the Lie algebra of the diffeomorphism group, i.e. the Lie algebra of vector fields on $\mathcal{E}$.

$\varepsilon \in L$

$$|\varepsilon|^2 = \int_{\mathcal{E}} (\varepsilon, \varepsilon) \text{ vol. form} = \int_{\mathcal{E}} (\varepsilon, \varepsilon)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{10}$$

$$\delta_\varepsilon g \in S^2(T^*) \tag{11}$$

[At right:] (variation of the metric in the direction ε).

[Below the expression:] — Lie derivative.



Uncertain reading. The crossed-out words below $\delta_\varepsilon g$ may read "vector fields."

Original source details: GELFAND-P001215

define the operator:

$$L = \Gamma(T) \xrightarrow{D} \Gamma(S^2(T^*)), \qquad D(\varepsilon) = \delta_\varepsilon g \tag{12}$$

[Below $\Gamma(T)$*:]* (vector fields). (g is fixed).



Uncertain reading. The overwritten exponent on S on the right of (12) reads 2.

Original source details: GELFAND-P001215

$$\|\varepsilon\|^2 = \int_{\mathcal{E}} (D\varepsilon, D\varepsilon) \text{ vol. form} = \int_{\mathcal{E}} (D\varepsilon, D\varepsilon)\sqrt{g}\, d\xi_1\, d\xi_2 \tag{13}$$

$$\frac{\text{vol. form corresponding to } \|\quad\|}{\text{vol. form corresponding to } |\quad|} = (\det \Delta)^{1/2}, \qquad \text{where } \Delta = D^*D \tag{14}$$

[In the right margin, beyond the double line:] Δ — Laplace operator on vector fields; $\Delta : \Gamma(T) \to \Gamma(T)$.

The ratio (14) is by definition the factor ν in (7)

$$\Gamma(S^2T) \subset \Gamma(\text{Hom}(T, T))$$

[Below Hom(T, T)*, brace:]* scalars $\oplus$ trace 0.

[At right:] (because of the metric); (identify S^2T, S^2T^* and embed in Hom(T, T)).

Corresponding decompositions of the sections:

$$\Gamma(S^2T) = \Gamma(S^2T)^c \oplus \Gamma(S^2T)^{\text{diff}} \tag{15}$$

[Below the first summand:] conformal changes of the metric.

[Below the second summand:] changes under the action of Lie derivatives.

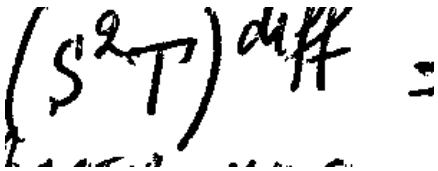
[crossed out]

$$(S^2T)^{\text{diff}} = T \otimes \Omega^{0,1} \tag{16}$$



Original source details: GELFAND-P001215

Uncertain reading. A mark before the left side of (16) is crossed out, possibly Γ or Φ.



Original source details: GELFAND-P001215

Uncertain reading. The superscript reads "diff."

[crossed out] It turns out that

$$D = \bar{\partial} : \Gamma(T) \longrightarrow \Gamma(T \otimes \Omega^{0,1}) \tag{17}$$



Original source details: GELFAND-P001215

Uncertain reading. An undeciphered mark before "It turns out that" is crossed out.

[The metric determines on $\mathcal{E}$ a complex structure, so it makes sense to write $\Omega^{0,1}$]

$$(S^2T)^{\text{diff}} \sim \text{-matrix } \begin{pmatrix} a & b \\ b & -a \end{pmatrix};$$

the meaning of (16) — the matrix is determined by its first row.

T — a holomorphic bundle, $\Omega^{0,1}$ — a 1-dimensional complex bundle.

on T acts the group of rotations.

S^2T — a 3-dimensional bundle, decompose it under the action of the rotation group.

GELFAND-P001216

[source number: 80]

Explanation of formula (14):

[Δ — the operator corresponding to quadratic form (13) relative to quadratic form (10);].

Incidentally:

$\delta_\varepsilon g$ — the Lie derivative of the metric g in the direction ε.

[A second way to define Δ on $\Gamma(T)$: identify T and T^* using the metric and take Δ on 1-forms. We get the same thing, since d (de Rham differential) and the Lie derivative are dual to one another].



Original source details: GELFAND-P001216

Uncertain reading. The last line in brackets reads “and the Lie derivative are dual to one another.”

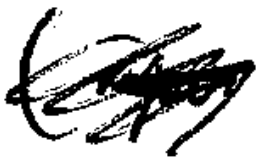
$S^2T = S^2T^*$ in the presence of the metric.

GELFAND-P001217

21/X–85. Gelfand Seminar.

Ryzhikova (A. Semenov’s student, in her 5th year).

An algorithm for checking any identity in the structure of subspaces [crossed out] of a finite-dimensional space.



Undeciphered passage. The crossed-out parenthesis before “finite-dimensional space” is undeciphered.

Original source details: GELFAND-P001217

We see which dimension suffices for checking [crossed out]



Uncertain reading. The crossed-out mark after “checking” may be 2 or a question mark.

Original source details: GELFAND-P001217

$$a(ab+c)=ab+ac \qquad \text{or} \qquad a(b+c)=ab+ac \quad \text{when } b\le a$$

[Below:] dimension 2 suffices.

General formula: it is enough to check in a space of dimension

$$n=\min[\max(d(p),d(q)),\max(k(p),k(q))]+1$$

p,q — 2 given terms

$d(p)$ — the number of additions in p (disjunction)

$k(p)$ — " — multiplications in p (conjunction)

In a space of fixed dimension (over $\mathbb{R}$) there is Tarski's algorithm.

Fuchs (following Feigin). Continuation on strings.

We need to evaluate the integral:

$$\int_{\mathcal{M}^c}(\operatorname{Det}\Delta)^{-d/2}\nu \tag{1}$$

Feigin.

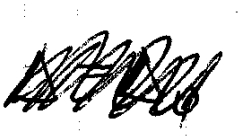
We need to compute the integral

$$\int_{\mathcal{M}_t}(\operatorname{Det}\Delta_0)^{-d/2}\mathcal{D}g=\int_{\mathcal{M}_t^c}\frac{\operatorname{Det}\Delta_1}{(\operatorname{Det}\Delta_0)^{d/2}}\mathcal{D}g, \tag{2}$$



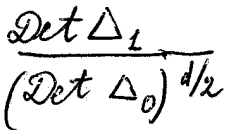
Original source details: GELFAND-P001217

Uncertain reading. In (2) the superscript on $\mathcal{M}_t$ under the first integral is crossed out, possibly c.



Original source details: GELFAND-P001217

Uncertain reading. The writing to the right of (2) is crossed out, possibly $\Delta=\Delta_0$ or $\Delta\neq\Delta_0$.



Original source details: GELFAND-P001217

Uncertain reading. The numerator reads $\operatorname{Det}\Delta_1$ with no visible exponent $1/2$.

$\mathcal{M}$ — the space of metrics, t — a point in the moduli space of curves of genus g

$\mathcal{M}_t$ — the space of metrics for the complex structure corresponding to modulus t.

GELFAND-P001218

[source number: 87]

$\mathcal{M}_t^c$ — the set of metrics [inserted above the line: (corresponding to the given modulus t)] of the form $e^{\varphi} g_0$, where g_0 — a metric of constant negative curvature $= -1$.



Original source details: GELFAND-P001218

Uncertain reading. The note below $\mathcal{M}_t^c$ appears to abbreviate "on the surface corresponding to t."

T — the tangent bundle on the curve (already equipped with a complex structure)

$$\bar{\partial} : \Gamma(T^j) \longrightarrow \Gamma(T^j \otimes \Omega^{0,1}), \qquad \Delta_j = \bar{\partial}^* \bar{\partial}$$



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Uncertain reading. A mark after T^j, before $\otimes$, is crossed out, probably a closing parenthesis.

In fact $\mathcal{M}_t^c$ [crossed out]; $\mathcal{M}_t^c \subset \mathcal{M}_t$ [crossed out]



Original source details: GELFAND-P001218

Uncertain reading. An undeciphered mark after the first $\mathcal{M}_t^c$ is crossed out.



Original source details: GELFAND-P001218

Uncertain reading. At right, the crossed-out expression may be $(\mathcal{M}^c \subset \mathcal{M}_t)$ or $(\mathcal{M}_t^c \subset \mathcal{M}_t)$.

(t is fixed).

Every metric in $\mathcal{M}_t$ can be (by the action of Diff)



Original source details: GELFAND-P001218

Uncertain reading. The sentence stops after "(by the action of Diff)."

The moduli space has dimension $3g - 3$.

The integral over it has yet to be evaluated (much work is devoted to it).



Original source details: GELFAND-P001218

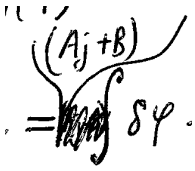
Undeciphered passage. The short crossed-out formula before the line about Det Δ_0 is undeciphered.

Det Δ_0 — is taken over the nonzero eigenvalues.

Det Δ_j = Det $\Delta_j(\varphi)$.

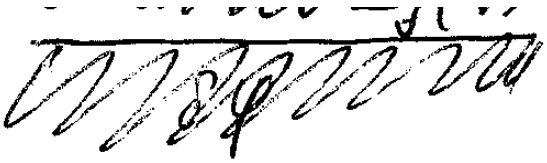
Next,

$$\delta \ln \mathrm{Det}\, \Delta_j(\varphi) = (Aj + B) \int_{\mathcal{E}} \delta\varphi \cdot \mathrm{Vol} + C(6j^2 + 6j + 1) \int_{\mathcal{E}} \delta\varphi \cdot K \cdot \mathrm{Vol} \tag{3}$$



Original source details: GELFAND-P001218

Uncertain reading. The original coefficient is crossed out. The insertion reads $(Aj + B)$, possibly $(A_j + B)$.



Original source details: GELFAND-P001218

Uncertain reading. The fraction bar and denominator $\delta\varphi$ under the left side are crossed out.

[Below the first Vol*, brace:]* volume form.

(j — "spin")

A, B, C — some constants; K — the Gaussian curvature of the surface.

$$j = 1 \Rightarrow 6j^2 + 6j + 1 = 13$$
$$j = 0 \Rightarrow 6j^2 + 6j + 1 = 1$$



Original source details: GELFAND-P001218

Uncertain reading. The result on the $j = 0$ line is overwritten and reads 1.

At $d = 26$ the anomaly cancels.

And then one has to integrate a constant (on the right-hand side of (2))

Corollary:

$$\ln \mathrm{Det}\, \Delta_j(\varphi) = A \int [\text{crossed out}]\, \varphi \cdot \partial\bar{\partial}\varphi + B \int e^{\varphi}\, \mathrm{Vol} + C \int \varphi\, \mathrm{Vol} + D \tag{4}$$



Original source details: GELFAND-P001218

Uncertain reading. The crossed-out expression immediately after the first integral sign in (4) may read $\partial\varphi \wedge \bar{\partial}\varphi$ or $\partial\varphi\, \bar{\partial}\varphi$.

(A, B, C, D — constants)

$\Rightarrow$ the calculation is equivalent to the Liouville equation (in (4) its Lagrangian)

Solving the Liouville equation is equivalent to computing

$$\int e^{\mathcal{L}_j(\varphi)} \mathcal{D}\varphi, \tag{5}$$

where $\mathcal{L}_j(\varphi)$ — left (= right) side of (4)
$\ln \operatorname{Det} \Delta_j$ is taken as the Lagrangian of the new theory. It is clear that

$$(2) = \int e^{\mathcal{L}_1(\varphi) - \frac{d}{2}\mathcal{L}_0(\varphi)} \mathcal{D}\varphi. \tag{6}$$